1. Item 6

> Identify and describe the mathematical aspects of a real-world problem situation including identifying the significant variables

This problem presents a real-world scenario (Ferris wheel motion). Students need to identify the mathematical elements involved:

- Key variables: rotation period (40 minutes), total angle (360°), time elapsed (30 minutes), starting position (point P)
- Mathematical relationships: uniform circular motion, proportional relationship between angle and time

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2. Item 4

> Read, decode and make sense of statements, questions, tasks, objects or images to create a model of the situation

Students need to:

- Understand the textual descriptions ("rotates at a constant speed," "one full rotation in 40 minutes," "starts at point P")
- Decode the labels in the diagram (points P, R, S and their relative positions)
- Establish a "time-angle-position" correspondence model

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3. Item 7

> Simplify or decompose a situation or problem in order to make it amenable to mathematical analysis

The problem simplifies the Ferris wheel motion by:

- Ignoring real-world factors like boarding, acceleration, or passenger movement
- Reducing it to a uniform circular motion model
- Translating the "position" question into an "angle" problem

4. Item 9

> Translate a problem into a standard mathematical representation or algorithm

Students need to translate the problem into a mathematical model:

$$\text{Rotation angle} = \frac{30}{40} \times 360^\circ = 270^\circ$$

and understand the resulting position after a 270° clockwise rotation from point P.

5. Item 5

> Recognize mathematical structure (including regularities, relationships, and patterns) in problems or situations

Students must recognize:

- The proportional relationship between time and angle
- One full rotation corresponds to 360°
- Points P, R, S approximately divide the circle into quadrants (each about 90°)

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6. Item 10

> Use mathematical tools (using appropriate variables, symbols, diagrams) to describe the mathematical structures and/or relationships in a problem

Students may need to:

- Sketch a simple circle diagram and mark angles
- Use proportional expressions to represent the time-angle relationship
- Use arrows to indicate rotation direction and final position

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7. Item 8

> Recognize aspects of a problem that correspond with known problems or mathematical concepts, facts or procedures

This is essentially a **proportional reasoning problem combined with circular motion**, relating to:

- Angle rotated per unit time
- Correspondence between fractions and circular proportions
- Judging direction of clockwise rotation

8. Item 1

> Select a mathematical description or a representation that describes a problem

Students must choose a "proportional model" or an "angular displacement model" to describe this motion problem.